

# The Capacity of a Fixed-Size Fast-Weight State: A Located Frontier That Grows Super-Linearly with Dimension

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Paper under double-blind review

## Abstract

How many independent bindings does a fixed-size fast-weight state hold before recall collapses? We measure this directly in a single-layer DeltaNet trained on a synthetic associative-recall grammar, grading recovery by exact continuous cosine similarity rather than argmax over a codebook, and we report three results and one provisioning law. First, at state dimension  $d_{\text{state}} = 64$  the held-out 4-hop recall frontier is a sharp transition, not a graceful decline: a logistic fit over the measured loads locates the mid-point at  $K/d_{\text{state}} = 0.5455$  (95% bootstrap CI  $[0.5385, 0.5513]$ ). Second, the ratio is not the law: the identical  $K/d_{\text{state}}$  window is flat at ceiling at  $d_{\text{state}} = 128$  (12 of 12 cells at  $h4 = 1.0$ ), the  $d_{\text{state}} = 80$  frontier sits at 0.6779 (CI  $[0.6683, 0.6867]$ ), excluding a pre-registered ratio-invariance band, and  $d_{\text{state}} = 96$  shows no transition anywhere through  $K/d_{\text{state}} = 0.9375$ . Between the two located frontiers, capacity in raw bindings grows as a two-point  $d^{1.97}$  exponent (CI-propagated range  $[1.86, 2.09]$ ): proportional to state bytes, not state width. Third, the leading single-variable account of the growth, anchor-table coherence, fails a direct test: frozen coherence injected at and above the collapsing scale’s own realized training band leaves recall at exact ceiling in 19 of 19 cells, under both concentrated and diffuse injection structures. Fixed-size states are the constant-memory endpoint of the KV-cache trade-off; these measurements give their capacity a located, pre-registered frontier.

## 1 Introduction

The memory cost of transformer inference grows with context: every decoded token appends to a key–value cache, and long-horizon reasoning workloads pay for that growth in high-bandwidth memory. One line of work manages the growing cache by evicting or compressing entries (Xiao et al., 2024; Zhang et al., 2023). A second line removes the growth altogether: linear-attention and fast-weight architectures carry a fixed  $d_{\text{state}} \times d_{\text{state}}$  matrix state that is updated in place as the sequence streams (Katharopoulos et al., 2020; Schlag et al., 2021; Yang et al., 2024b; Gu & Dao, 2023). A fixed-size state is the constant-memory endpoint of the cache trade-off, and it turns a scheduling question into a provisioning question: for a state of a given byte size, how many independent bindings can it be trusted to hold, and how does that number move when the byte budget changes?

Existing answers do not settle the provisioning question. Classical linear associative memories have information-theoretic ceilings near  $K \approx d$  for  $K$  independent bindings under exact linear reads (Kohonen, 1972; Anderson, 1972), and recent theory sharpens such thresholds for linear associative recall (Barnfield et al., 2026). Under argmax decoding the accounting changes in kind: a rank-one state can recover on the order of  $d$  associations (Nichani et al., 2025), so any capacity number is meaningful only relative to its readout. Descriptive studies track the effective rank of trained linear-attention states at scale (Nazari & Rusch, 2026; Sun et al., 2026). What none of these supply is a located frontier for a trained delta-rule memory under an exact continuous readout: the load at which recall collapses, measured with a confidence interval, and its scaling in  $d_{\text{state}}$ .

This paper supplies that frontier on a controlled testbed, with three results and one provisioning law.

First, the transition, located. At  $d_{\text{state}} = 64$ , held-out 4-hop compositional recall does not decline gracefully with load: it transitions sharply, and a logistic fit over the measured loads places the midpoint at  $x_0 = K/d_{\text{state}} = 0.5455$  (95% seed-level bootstrap CI  $[0.5385, 0.5513]$ , width 0.0127), a roughly  $20\times$  tightening over the prior bracket between the two flanking archived loads.

Second, the ratio is not the law. The identical  $K/d_{\text{state}}$  window that collapses at  $d_{\text{state}} = 64$  is flat at exact ceiling at  $d_{\text{state}} = 128$ :  $h4 = 1.0$  in 12 of 12 cells. The frontier at  $d_{\text{state}} = 80$  sits at  $x_0 = 0.6779$  (CI  $[0.6683, 0.6867]$ ), excluding a pre-registered ratio-invariance band  $[0.4745, 0.6165]$  entirely, and  $d_{\text{state}} = 96$  shows no transition anywhere in a window reaching  $K/d_{\text{state}} = 0.9375$ . Between the two located frontiers, capacity in raw bindings grows as  $d^{1.97}$  (CI-propagated two-point range  $[1.86, 2.09]$ ): consistent with proportional to state bytes ( $d^2$ ), excluding proportional to state width ( $d^1$ ). A load rule calibrated at one scale therefore mis-provisions another; the safe load ratio itself grows with  $d_{\text{state}}$ .

Third, the leading confound, tested and exonerated. The collapsing and non-collapsing scales differ in more than size: with  $n = 107$  entities, the  $d = 64$  anchor table is forced non-orthogonal by the Welch bound while the  $d = 128$  table is exactly orthogonal, making table coherence the most natural single-variable account. Injecting controlled, frozen coherence into the flat  $d = 128$  geometry, at and above the  $d = 64$  table’s own realized training band, leaves recall at exact ceiling in 19 of 19 cells under both a concentrated (rank-4) and a diffuse (rank-48) injection structure. Scalar table coherence, at matched dose and matched geometry, does not reproduce the transition.

A binding count as a reasoning-memory proxy. Multi-step reasoning traces carry intermediate results as sub-goals or variable bindings that later steps must retrieve: a chain that derives seven intermediate quantities before its answer holds seven bindings in working memory at once, the same simultaneous-binding load this testbed isolates directly. A fixed-size fast-weight state that keeps intermediate reasoning state in place, rather than replaying it through a growing key-value cache, inherits exactly the load limit measured here: the located frontier converts a state’s byte budget into a concrete bound on how many sub-goals it can hold before recall degrades. The synthetic grammar does not model any particular reasoning task, but the binding-count abstraction it isolates is the quantity a constant-memory reasoning architecture would need to provision.

Every wave reported here was pre-registered before its cells ran, with outcome semantics fixed in advance, including the rule that a degenerate sigmoid fit on flat data must be disclosed as such and never quoted as a located transition; instrument saturation is checked at every wave rather than assumed. The scope is disclosed up front: a single-head, single-layer DeltaNet, a 107-entity synthetic grammar, sub-million-parameter models, and  $d_{\text{state}} \in \{64, 80, 96, 128\}$ . No claim is made about production-scale models; Section 7 states the boundaries.

## 2 Testbed and Measurement Process

Architecture and task. All experiments use a vanilla single-head, single-layer DeltaNet with the production `chunk_delta_rule` kernel (Yang et al., 2024b), whose state update is  $S_t = S_{t-1} (I - \beta_t k_t k_t^\top) + \beta_t v_t k_t^\top$  (Schlag et al., 2021). The task is synthetic-grammar associative recall over a pool of 107 trainable entities: within an episode, each queried entity must be bound to a value through the delta rule and recovered later through multi-hop composition. The model width  $d_{\text{model}} = 256$  is held fixed across the entire  $d_{\text{state}}$  sweep, so the state dimension is the sole varied architectural axis in every comparison below. Models are trained for 20,000 steps per cell; each (load, seed) pair is one cell.

Readout. Recovery is graded by exact continuous cosine similarity against the true value vector at threshold 0.9, written `rec@0.9`, never by argmax over a codebook. The distinction is load-bearing: under argmax decoding a rank-one state can recover on the order of  $d$  associations (Nichani et al., 2025), so a capacity measurement that permits codebook lookup measures the codebook, not the state. The paper’s single load-bearing quantity is  $h4$ , the

rec@0.9 fraction at hop depth 4, with the queried entities drawn from the same 107-entity pool used in training; the held-out axis is hop depth (models train on hop depths 1–3 and are evaluated at depth 4), not entity identity. Every reported recovery is therefore a held-out-depth compositional generalization over familiar entities, not a memorized pair; it is not a test of generalization to unfamiliar entities, which Section 7 reports separately and finds does not hold on this testbed.

The load parameter.  $K$  is the number of independent entity bindings the state must support simultaneously within one episode;  $K/d_{\text{state}}$  is the capacity ratio. The frontier statistic  $x_0$  is the fitted midpoint of a logistic  $h4(x) = L/(1 + \exp((x - x_0)/w))$  in  $x = K/d_{\text{state}}$ : the load at which recall crosses half of its ceiling.  $x_0$  is a well-defined transition location, not the largest load with perfect recall; the fitted width  $w$  reports how sharp the crossing is.

The anchor table and coherence. A per-entity key-anchoring table (established in companion work on this testbed as improving held-out-depth recovery, and treated here as part of the fixed architecture) is the object whose geometry differs across scales. Its coherence is  $\max|\cos|$  over the table’s 107 row vectors. The Welch bound (Welch, 1974) forces this strictly positive whenever  $n = 107 > d_{\text{state}}$ , as at  $d_{\text{state}} = 64$ , and permits exact orthogonality when  $n \leq d_{\text{state}}$ , as at  $d_{\text{state}} = 128$  where the constructed table measures  $\max|\cos| = 0.000000$ . This single structural difference motivates the dose-response test of Section 4.

Anchor-table byte footprint. The anchor table is  $n \times d_{\text{state}}$  at fp32,  $428 \cdot d_{\text{state}}$  bytes: 26.75 KiB at  $d_{\text{state}} = 64$ , exceeding the 16 KiB state itself by 67%, and 53.5 KiB at  $d_{\text{state}} = 128$ , 84% of the 64 KiB state. The table’s footprint is  $O(n \cdot d_{\text{state}})$ , linear in state dimension, against the state’s own  $O(d_{\text{state}}^2)$ ; the ratio of table to state bytes is  $n/d_{\text{state}}$ , which is why the table dominates at the smallest tested scale and falls below the state’s own size by  $d_{\text{state}} = 128$ . Because the table’s size is fixed by the entity count  $n$  and does not scale with the load  $K$ , we exclude it from the bindings-per-KiB accounting of Table 1, which reports marginal capacity per byte of load-bearing state; a deployment budget that provisions this architecture must add the table’s bytes to the state’s.

Pre-registration and instrument discipline. Each wave’s hypothesis and outcome semantics were fixed before its cells ran. Three rules matter downstream. (i) Degenerate-fit disclosure: a sigmoid fitted to flat-at-ceiling data is degenerate by construction; the registered rule is to report the degenerate bootstrap fraction and refuse to quote the point estimate as a located transition. (ii) Instrument saturation is checked, not assumed: hop-21 recovery, a far harder extrapolation than  $h4$ , is recorded at every wave; a flat  $h4$  result is reportable only when hop-21 is not uniformly 1.0, ruling out a globally ceilinged readout. (iii) Admission gates: every cell passes mechanical admissibility checks (convergence of the eval-time Newton–Schulz block, value-salvage floors, finite loss) before its  $h4$  enters any fit; Section 5 reports the one place where the gate itself, not the measurement, required recalibration.

### 3 The Transition, Located and Dissolved

#### 3.1 A Sharp Transition at $d_{\text{state}} = 64$

At  $d_{\text{state}} = 64$ , seed-mean  $h4$  over the measured loads falls  $1.000 (K=16) \rightarrow 0.6669 (K=32) \rightarrow 0.5676 (K=34) \rightarrow 0.3316 (K=38) \rightarrow 0.1177 (K=42) \rightarrow 0.0434 (K=46) \rightarrow 0.0215 (K=48)$ : a collapse concentrated in a narrow band of load, not a graceful decline (Figure 1, left). A logistic fit with a 4,000-trial seed-level parametric bootstrap (0 of 4,000 fits degenerate) locates the midpoint at  $x_0 = 0.5455$  (95% CI  $[0.5385, 0.5513]$ , width 0.0127) with characteristic width  $w = 0.0597$  (CI  $[0.0557, 0.0642]$ ); the band  $x_0 \pm w$  spans  $K/d_{\text{state}} \approx [0.486, 0.605]$ , or  $K \approx 31\text{--}39$  raw bindings. Before this wave, the transition was known only to lie somewhere between the two archived flanking loads, a bracket 0.25 wide in ratio terms; the fit tightens that bracket by a factor of roughly 20. This is a statement about where the transition sits, not why it occurs; Sections 4 and 5 test candidate mechanisms directly.

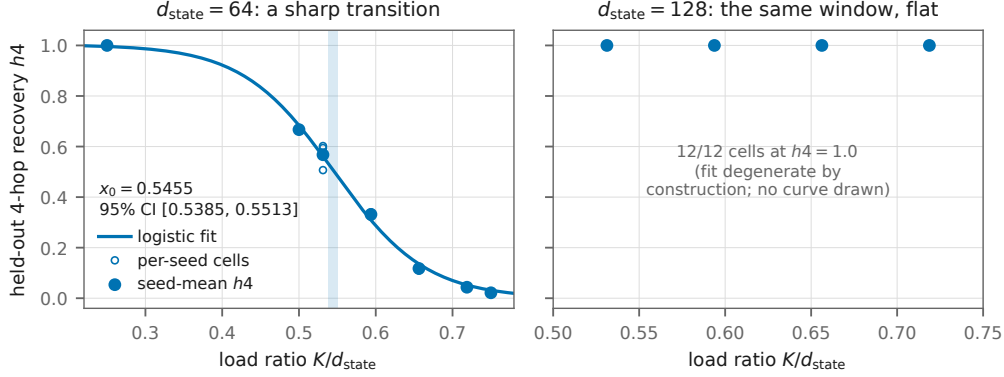


Figure 1: Held-out 4-hop recovery  $h_4$  (rec@0.9, exact continuous cosine, never argmax) versus load ratio  $K/d_{\text{state}}$  in a single-layer DeltaNet on the 107-entity recall grammar. Left: at  $d_{\text{state}} = 64$ , seed-mean recovery over the measured loads ( $K = 16$  saturation anchor, archived  $K = 32/48$ , new  $K = 34/38/42/46$  at three seeds each; open circles are per-seed cells) transitions sharply; the logistic fit locates the midpoint at  $x_0 = 0.5455$  with 95% bootstrap CI  $[0.5385, 0.5513]$  (shaded band). Right: the identical  $K/d_{\text{state}}$  window re-run at  $d_{\text{state}} = 128$  ( $K = 68/76/84/92$ , three seeds each) is flat at exact ceiling,  $h_4 = 1.0$  in 12 of 12 cells; the sigmoid fit on this flat data is degenerate by construction (bootstrap degenerate fraction 1.0) and no curve is drawn, per the pre-registered disclosure rule. Takeaway: the same load ratio that collapses recall at  $d_{\text{state}} = 64$  is loss-free at  $d_{\text{state}} = 128$ .

### 3.2 The Same Window, Flat at $d_{\text{state}} = 128$

Re-running the identical  $K/d_{\text{state}}$  window at  $d_{\text{state}} = 128$ , with matched loads  $K = 68, 76, 84, 92$  at three seeds each, produces  $h_4 = 1.0$  in every one of the 12 cells (Figure 1, right). Per the pre-registered outcome semantics this is the strong form of a shifted transition: not a moved midpoint, but a transition that exits the measured window entirely. At  $d_{\text{state}} = 128$  the state supports at least 92 of the 107 entities through held-out 4-hop composition, in the exact ratio band where  $d_{\text{state}} = 64$  had collapsed from 0.667 to 0.022.

Two checks rule out a broken or saturated instrument. Hop-21 recovery, the far harder extrapolation, spans 0.9900–1.0 across the 12 cells rather than sitting uniformly at 1.0, and the state’s measured effective rank tracks the load almost exactly (at  $K=84$ : 83.7/83.6/83.4 across the three seeds). The state is using its permitted rank, and the readout retains headroom; the flat  $h_4$  row is a real ceiling of the measured quantity. The immediate consequence: the capacity ratio  $K/d_{\text{state}}$  alone does not determine where recall collapses.

## 4 The Coherence Confound, Exonerated Twice

The two scales of Section 3 differ in more than raw size. With  $n = 107$  entities, the  $d_{\text{state}} = 64$  anchor table is forced non-orthogonal by the Welch bound: it measures  $\max|\cos| = 0.2842$  at construction and drifts under ordinary training to a final-checkpoint band of per-load means  $[0.373, 0.385]$  (recomputed from the archived cliff cells). The  $d_{\text{state}} = 128$  table is exactly orthogonal at construction. Table coherence is therefore the single most natural remaining single-variable account of why the transition vanished, and it is testable directly rather than by inference across scales.

Design. Holding  $d_{\text{state}} = 128$ ,  $n = 107$ , and  $K = 68$  fixed at exactly the geometry measured flat in Section 3.2, we inject controlled coherence into the anchor table, freeze the table, and sweep the dose: 0.130, 0.284 (matching the  $d = 64$  construction value), and 0.40, the last exceeding the  $d = 64$  table’s entire realized training band. The injection structure is a registered co-primary axis: a concentrated variant confines the injected overlap to a rank-4 subspace, and a diffuse variant spreads the same scalar dose across a rank-48 subspace.

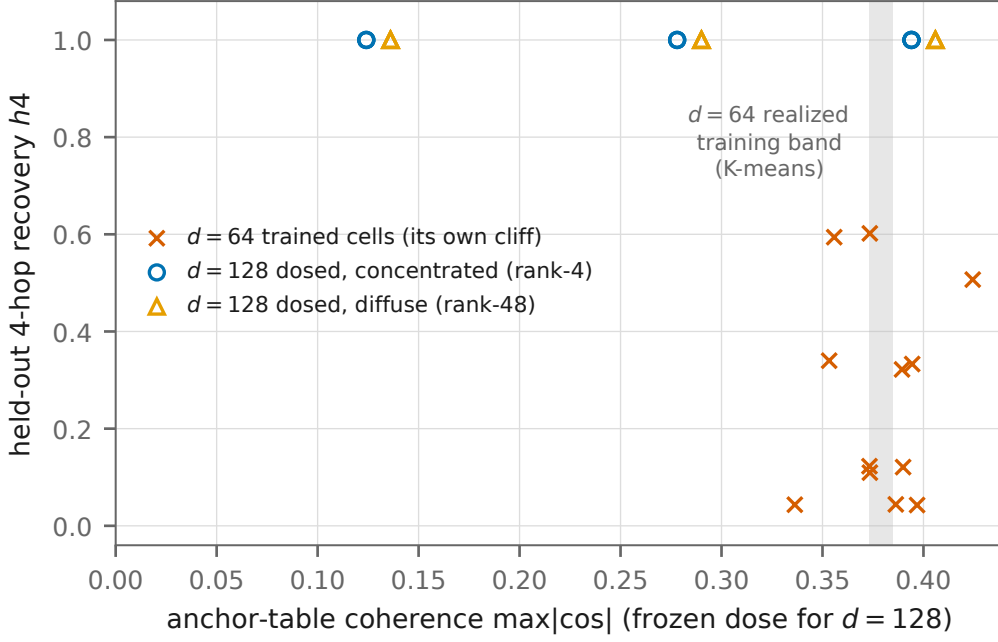


Figure 2: Coherence dose-response at  $d_{\text{state}} = 128$ ,  $K = 68$ , the geometry Section 3.2 measured flat. Blue circles: frozen coherence injected under a concentrated (rank-4) structure; orange triangles: the same doses under a diffuse (rank-48) structure; the two marker families are offset horizontally by  $\pm 0.006$  for visibility because all 19 cells coincide at  $h4 = 1.0$ . Vermillion crosses: the  $d_{\text{state}} = 64$  cliff cells plotted against their own final-checkpoint table coherence, collapsing across the same coherence range. Grey band: the range of  $d_{\text{state}} = 64$  per-load mean final coherence,  $[0.373, 0.385]$ . Takeaway: directly injected coherence at and above the collapsing scale’s own training band does not move recall off ceiling.

Freezing is verified, not assumed: the pre-launch gate requires the table’s gradient path to be severed, and the recorded coherence is bit-identical across all 10 checkpoints of every dosed cell (a single distinct value per cell across the full 20,000-step run).

Result. Recall stays at exact ceiling everywhere:  $h4 = 1.0$  in 10 of 10 concentrated cells and 9 of 9 diffuse cells, at every dose, with no exception (Figure 2). Hop-21 recovery across the 19 dosed cells spans 0.9963–1.0, not uniformly 1.0, so the flat  $h4$  row is a genuine ceiling of the measured quantity rather than a saturated instrument. Per the pre-registered outcome semantics this is the strongest exoneration the design can produce: directly injected anchor-table coherence, at matched  $K/d_{\text{state}}$  and matched entity count, held frozen for the full run, does not reproduce the  $d_{\text{state}} = 64$  collapse under either injection structure. What this does not rule out is stated in Section 7: a coherence profile that co-evolves with training, rather than being frozen at initialization, remains a disclosed open variant.

## 5 The Capacity Frontier Grows Super-Linearly

If the transition of Section 3.1 were a property of the ratio  $K/d_{\text{state}}$ , its location would be invariant across scales. The invariance band  $[0.4745, 0.6165]$ , centered on the  $d_{\text{state}} = 64$  midpoint, was registered before any larger-scale cell ran.

The frontier at  $d_{\text{state}} = 80$ . With loads  $K \in \{20, 43, 48, 53, 58\}$  at three seeds each, escalated to five seeds at the two mid-transition loads, the refit locates  $x_0(80) = 0.6779$  (95% CI  $[0.6683, 0.6867]$ ,  $w = 0.0479$ , 0 of 4,000 bootstrap fits degenerate). The CI excludes the

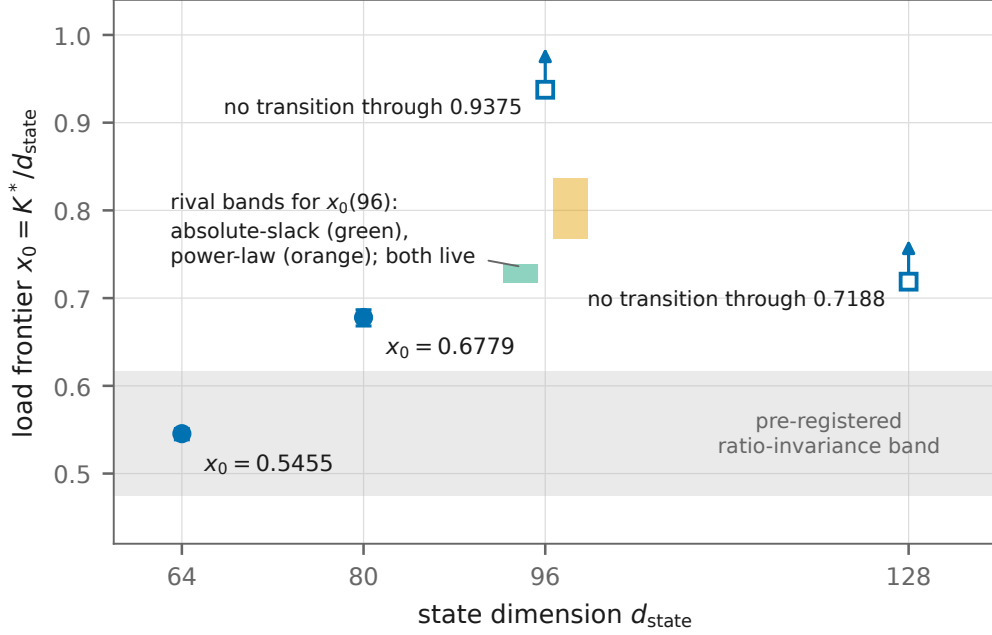


Figure 3: The load frontier  $x_0 = K^*/d_{\text{state}}$  versus state dimension. Filled circles with error bars: located transition midpoints with 95% bootstrap CIs at  $d_{\text{state}} = 64$  and  $80$ . Open squares with arrows: window-limited lower bounds where no transition appears anywhere in the measured window ( $d_{\text{state}} = 96$  through  $K/d = 0.9375$ ;  $d_{\text{state}} = 128$  through  $0.71875$ ). Grey band: the pre-registered ratio-invariance band  $[0.4745, 0.6165]$ , which the  $d_{\text{state}} = 80$  CI excludes entirely. Green and orange patches at  $d_{\text{state}} = 96$ : the two registered rival predictions for  $x_0(96)$  (absolute-slack  $[0.718, 0.739]$ ; power-law  $[0.768, 0.837]$ ), neither excluded by the degenerate fit. Takeaway: the safe load ratio itself grows with state dimension.

invariance band entirely, on the high side: in ratio terms the frontier moved up by  $+0.13$  as  $d_{\text{state}}$  grew from  $64$  to  $80$ . Ratio invariance is refuted, in the direction of more usable load per dimension at larger scale (Figure 3).

No transition through  $K/d = 0.9375$  at  $d_{\text{state}} = 96$ . The registered  $d_{\text{state}} = 96$  grid ( $K = 24/51/57/63/69$ , three seeds) is flat near ceiling across  $K/d \in [0.25, 0.71875]$ , with per-load means  $1.0/1.0/0.9974/0.9805/0.9839$ ; its sigmoid fit is degenerate (98.5% of bootstrap resamples) and is disclosed as such, never quoted as a location. A wider grid ( $K = 69/72/78/84/90$ , three seeds) initially returned 11 of its 12 new cells inadmissible at the standing admission gate. An independent diagnostic instrument traced the failures to a tolerance miscalibration in the gate’s convergence check, not to the cells: every one of the 4,608 flagged evaluation episodes resolves by  $n_{\text{iter}} \leq 28$  against the production setting of 20. Only the admission flag was recalibrated (11 cells flipped); no  $h4$  measurement changed and no cell re-ran. A thirteenth examined cell, a pre-existing  $K=69$  seed from an earlier wave with the identical failing signature, was deliberately excluded per the recalibration’s declared 11-cell scope; including it would add a fourth  $K=69$  seed and shift that load’s mean from  $0.9592$  to  $0.9423$ , changing no conclusion. On the unlocked grid the per-load means are  $0.9592/0.9216/0.9326/0.9581/1.0000$  across  $K/d \in [0.71875, 0.9375]$ : no monotonic collapse anywhere in the window, a non-monotonic dip-and-recovery instead, and a fit that is 100% bootstrap-degenerate from genuine per-seed scatter (the noisiest load,  $K=72$ , spans  $0.8426$ – $0.9904$  across seeds). The supportable summary is a window-limited bound: at  $d_{\text{state}} = 96$ , no transition through  $K/d = 0.9375$ , with the fine structure near ceiling left unresolved (Section 7).

| $d_{\text{state}}$ | state bytes (fp32) | frontier $x_0$                     | $K^*$ (bindings) | bindings/KiB  |
|--------------------|--------------------|------------------------------------|------------------|---------------|
| 64                 | 16,384 (16 KiB)    | 0.5455 (CI [0.5385, 0.5513])       | 34.9             | 2.18          |
| 80                 | 25,600 (25 KiB)    | 0.6779 (CI [0.6683, 0.6867])       | 54.2             | 2.17          |
| 96                 | 36,864 (36 KiB)    | $> 0.9375$ (window-limited bound)  | $\geq 90^+$      | $\geq 2.50^+$ |
| 128                | 65,536 (64 KiB)    | $> 0.71875$ (window-limited bound) | $\geq 92$        | $\geq 1.44$   |

Table 1: Provisioning arithmetic for a  $d \times d$  delta-rule state on this testbed.  $K^* = x_0 \cdot d_{\text{state}}$  is the located half-recovery load in raw bindings (rows 1–2) or the largest load measured at or near ceiling within the tested window (rows 3–4, bounds only). Bindings per KiB is  $K^*$  divided by state size at 32-bit precision. Between the two located rows, capacity per dimension rises ( $x_0$ : 0.5455  $\rightarrow$  0.6779) while capacity per byte is approximately conserved (2.18  $\rightarrow$  2.17 bindings/KiB). The  $d_{\text{state}} = 128$  row’s bound is loose because its measured window stops at  $K/d = 0.71875$ .  $^+$ Rests on a three-seed  $K=90$  reading ( $h4 = 1.0000$  in each seed); a fresh-seed replication attempt did not confirm the ceiling and was itself inadmissible at the recalibrated gate (Section 7).

Scaling in bytes, not width. Writing  $K^* = x_0 \cdot d_{\text{state}}$  for the located midpoint load in raw bindings, the two located frontiers give  $K^* = 34.9$  at  $d_{\text{state}} = 64$  and  $K^* = 54.2$  at  $d_{\text{state}} = 80$ . The two-point growth exponent is  $p = \ln(54.23/34.91)/\ln(80/64) = 1.97$ , with CI-propagated range [1.86, 2.09] from the two fits’ bootstrap intervals: consistent with  $p = 2$ , capacity proportional to state bytes ( $4d^2$  at fp32), and excluding  $p = 1$ , capacity proportional to state width. Table 1 states the same arithmetic per byte. Two registered functional forms for the growth remain live and undiscriminated at  $d_{\text{state}} = 96$ , where the fit is degenerate: an absolute-slack account predicting  $x_0(96) \in [0.718, 0.739]$  and a power-law account predicting [0.768, 0.837]. Both bands sit inside the flat measured window, so neither is excluded; discriminating them is registered follow-on work, not claimed here.

For a deployment engineer the reading is direct: over the measured range, spending bytes on a larger fast-weight state buys bindings at an approximately constant per-byte rate, and the safe load ratio tuned at a small state under-provisions a larger one by a CI-excluding margin.

## 6 Related Work

Fast-weight and linear-attention architectures. Katharopoulos et al. (2020) identify linear attention with a recurrent state; Schlag et al. (2021) connect it to fast weight programmers and introduce the delta-rule update this testbed trains; Yang et al. (2024b) give the parallel kernel used here. DeltaProduct (Siems et al., 2025), negative-eigenvalue variants (Grazzi et al., 2025), and gated delta rules (Yang et al., 2024a) extend the family’s expressivity, and matrix-state RNNs push it to language-model scale (Mishra et al., 2026); state-space models make the same constant-memory trade (Gu & Dao, 2023), with known expressivity limits (Merrill et al., 2024). Titans (Behrouz et al., 2025) and Infini-attention (Munkhdalai et al., 2024) add a bounded, test-time-updated memory module alongside or in place of attention, targeting the same fixed-memory budget this paper measures. None of these works locates a trained capacity transition or measures its movement with state dimension, which is this paper’s object.

Capacity of associative memories. Classical correlation-matrix memories establish the  $K \lesssim d$  regime for linear reads (Kohonen, 1972; Anderson, 1972); Barnfield et al. (2026) prove sharp thresholds for linear associative recall across retrieval rules. Nichani et al. (2025) show that under argmax decoding a rank-one state recovers on the order of  $d$  associations: the readout, not the state alone, sets the accounting. This paper’s contribution is empirical and complementary: a located frontier for a trained delta-rule memory under an exact continuous readout, the regime where within-state capacity is load-bearing.

Memory management for long-context inference. KV-cache eviction and compression manage a memory footprint that grows with context (Xiao et al., 2024; Zhang et al., 2023), and

recall-throughput trade-offs in efficient architectures are measured by Arora et al. (2023) and Arora et al. (2024), whose linear-attention models are architecturally adjacent to the single-head DeltaNet measured here. This paper addresses the fixed-size endpoint of that trade-off: not how much a growing cache can be shrunk, but how much a state of constant byte size can be trusted to hold.

Rank in trained linear-attention states. Nazari & Rusch (2026) and Sun et al. (2026) describe effective-rank dynamics of linear-attention states in trained LLMs. Those are descriptive accounts of what rank trained states occupy; the present work measures where task recall collapses as bound load approaches the state’s dimension, with pre-registered fits and a causal dose-response test on the leading confound.

## 7 Limitations

**Scope.** All results use a single-head, single-layer DeltaNet, one synthetic 107-entity recall grammar, sub-million-parameter models, and  $d_{\text{state}} \in \{64, 80, 96, 128\}$  ( $d_{\text{state}} = 32$  is excluded by a kernel-level failure in the underlying library, measured independently of this program). No claim transfers to production-scale gated variants (Siems et al., 2025; Grazzi et al., 2025), to matrix-state models at language-model scale (Mishra et al., 2026), or to natural language. The per-byte conservation reading of Table 1 rests on two located points plus one in-window bound; it is a measured regularity over  $d_{\text{state}} \in [64, 96]$ , not a law.

The held-out axis is hop depth, not entity identity. Every  $h4$  measurement in this paper draws its queried entities from the same 107-entity pool used in training; the held-out axis is hop depth alone (Section 2). A separate disjoint-entity-pool diagnostic, recorded at every checkpoint under a pre-registered scope limit (the anchoring mechanism claims no held-out-entity gains, and held-out entities bypass the anchor blend by construction), collapses at hop depth 2 and above: recovered fraction at threshold 0.9 is 1.0 at hop 1 in all 24 archived cells behind Figure 1, and at hop depths 2 and 3 it reads exactly 0.0 in 23 of those 24 cells (the remaining cell reads 0.0002), while the same checkpoints’ trained-pool readouts stay at ceiling at hops 1 and 2 and remain well above the held-out collapse at hop 3, where they decline under the highest tested loads (as low as 0.56 at  $d_{\text{state}} = 64$ ,  $K=46$ ). The located frontier and its scaling with  $d_{\text{state}}$  are therefore claims about capacity for entities the model has trained on, not about generalization to unfamiliar entities; the latter is a substantially harder, unresolved question on this testbed.

The frontier’s functional form is open. Four dimensions constrain the growth of  $x_0(d)$  but do not pin its form: the two located fits refute ratio invariance, yet the absolute-slack and power-law accounts both remain live at  $d_{\text{state}} = 96$ , where the fit is 100% bootstrap-degenerate from real per-seed scatter. Discriminating them requires either variance reduction at the noisiest loads or a window that brackets the  $d_{\text{state}} = 96$  transition from above, and a registered analysis found the present grid underpowered for that discrimination; we therefore make no functional-form claim.

Near-ceiling fine structure at  $d_{\text{state}} = 96$  is instrument-limited. The unlocked grid’s non-monotonic dip-and-recovery pattern, including the exact-ceiling reading at  $K=90$ , sits close to the admission instrument’s own resolution. A registered two-cell fresh-seed diagnostic aimed at this fine structure routed to its pre-registered degenerate outcome: neither cell produced an admissible reading at the recalibrated gate, and the failing admissibility leg differed from the one the original grid failed, so the diagnostic’s adjudication triggers were never licensed to fire. One of the two cells, at  $K=90$ , read  $h4 = 0.9725$  before its inadmissibility was determined; the archived  $K=90$  ceiling ( $h4 = 1.0000$  in each of three seeds, Table 1) did not replicate at this fresh seed. Per the diagnostic’s own pre-registered routing this reading cannot be promoted to a positive finding, because the cell is inadmissible, but we disclose it here as a caveat that discounts confidence in the archived  $K=90$  reading. We therefore scope the  $d_{\text{state}} = 96$  claim strictly as the absence of a monotonic collapse through  $K/d = 0.9375$ , and we treat the per-load ordering near ceiling as unresolved. The admission

gate episode itself (Section 5) cuts both ways: it shows the measurement pipeline can manufacture an apparent collapse, and it is disclosed precisely so that the reported transitions are read against that risk; the located  $d_{\text{state}} \in \{64, 80\}$  fits pass their gates without any recalibration.

The dose test bounds the tested variable only. The exoneration of Section 4 covers scalar coherence dose under two frozen injection structures (rank-4 and rank-48). It does not cover coherence that co-evolves with training: a non-frozen  $d_{\text{state}} = 128$  reference point whose table coherence drifted to  $[0.127, 0.202]$  under ordinary training also recovers at  $h4 = 1.0$ , but such a point confounds the coherence level with how the level was reached, and is reported as context, not as part of the controlled sweep. Mechanistic accounts below the level of the disclosed comparison axes (for example, optimization dynamics through the larger eval-time Newton-Schulz block) are untested here.

One architecture family. Whether the same three-part structure, a sharp transition, its dissolution with scale, and a coherence exoneration, appears in other fast-weight update rules or state-space families is untested; the measurement recipe (pre-registered logistic localization, degenerate-fit disclosure, saturation checks, admission gates) is the transferable artifact.

## 8 Conclusion

A fixed-size delta-rule state, the constant-memory endpoint of the long-context cache trade-off, has a measurable capacity frontier, and on this testbed that frontier is now located rather than bracketed: recall collapses in a sharp transition at  $K/d_{\text{state}} = 0.5455$  at  $d_{\text{state}} = 64$ , the same ratio window is loss-free at  $d_{\text{state}} = 128$ , the  $d_{\text{state}} = 80$  frontier excludes ratio invariance, and capacity in raw bindings grows with dimension at an exponent consistent with state bytes and inconsistent with state width. The most natural single-variable account of the growth, anchor-table coherence, fails a frozen dose-response test under two injection structures. For provisioning fixed-size memories, the practical rule is that safe load ratios do not transfer across state sizes. For the science of fast-weight memory, the residual question is now precisely posed: something other than the load ratio governs the frontier’s location, and scalar table coherence is not it.

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